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Nonnutritive Sweetener Consumption, Metabolic Risk Factors, and Inflammatory Biomarkers Among Adults in the Cancer Prevention Study-3 Diet Assessment Sub-Study

Allison C Sylvetsky [1,*], Ying Wang [2], Ananya G Reddy [2], Caroline Y Um [2], Rebecca A Hodge [2], Cari Lichtman [2], Diane Mitchell [3], Anuj Nanavati [2], Michael Pollak [4], Ye Wang [4], Alpa V Patel [2], Marjorie L McCullough [2]

[1] *Department of Exercise and Nutrition Sciences, Milken Institute School of Public Health, The George Washington University, Washington, DC, United States;* [2] *Department of Population Science, American Cancer Society, Atlanta, GA, United States;* [3] *Institute for Advancing Health through Agriculture, Texas A & M University, College Station, TX, United States;* [4] *Pollak Assay Lab, Jewish General Hospital, McGill University, Montreal, Quebec, Canada*

A B S T R A C T

Background: Nonnutritive sweeteners (NNSs) are widely used to replace added sugars, yet their role in metabolic health and chronic disease prevention is debated.

Objective: The objective of this study was to examine associations between NNS consumption, metabolic risk factors, and inflammatory biomarkers.

Methods: This cross-sectional analysis included 624 adults in the American Cancer Society's Cancer Prevention Study-3 Diet Assessment Substudy (DAS). Consumption of NNS, including aspartame, saccharin, sucralose, and acesulfame-potassium, was estimated using the mean quantities reported in 6 24-h dietary recalls over 1 y. Fasting insulin, C-peptide, hemoglobin A1c (HbA1c), leptin, adiponectin, C-reactive protein (CRP), tumor necrosis factor-alpha (TNF-α), interleukin-6 (IL-6), and interleukin-10 (IL-10) were measured in fasting blood samples collected twice, 6 mo apart. Multivariable linear regression was used to examine associations between NNS consumption and the mean levels of each metabolic or inflammatory biomarker. Base models were adjusted for age, sex, race, education, smoking, and physical activity; full models were further adjusted for body mass index (BMI), diet quality (Healthy Eating Index 2020), and energy intake.

Results: More than half (55%) of participants reported consuming NNS (mean daily NNS consumption 7, 38, and 221 mg across tertiles). NNS consumption was positively associated with leptin (*P*-trend = 0.0006) and CRP (*P*-trend = 0.02), but associations were attenuated after adjustment for BMI, diet quality, and energy intake. NNS consumption was not associated with insulin, C-peptide, HbA1c, adiponectin, TNF-α, or IL-10. In analyses stratified by BMI, NNS consumption was positively associated with IL-6 among participants with BMI $\geq$25kg/m^2 but not BMI $<$25kg/m^2.

Conclusions: Findings in the full sample were null after adjustment for energy intake and BMI, but NNS consumption was positively associated with IL-6 among participants with overweight or obesity. Investigation of mechanisms through which NNS consumption may impact inflammatory pathways is warranted.

Keywords: artificial sweetener, cardiometabolic disease, diet soda, sugar, diabetes, obesity, cardiovascular disease, cancer

Introduction

Nonnutritive sweeteners (NNSs) such as aspartame, acesulfame-potassium, saccharin, and sucralose are commonly used to replace added sugars in food and beverage products. However, their role in human health is a topic of ongoing debate. In May 2023, the WHO issued a recommendation stating that NNS should not be used for weight control or to reduce

Abbreviations: ACS, American Cancer Society; CPS-3, Cancer Prevention Study-3; CRP, C-reactive protein; CV, Coefficient of variation; DAS, Diet Assessment Substudy; HbA1c, hemoglobin A1c; ICC, intraclass correlation; NNS, nonnutritive sweetener.

* Corresponding author. *E-mail address:* asylvets@gwu.edu (A.C. Sylvetsky).

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noncommunicable disease risk [1]. In July 2023, the International Agency for Research on Cancer classified aspartame—one of the most commonly used NNS—as a "possible carcinogen," whereas the Joint FAO/WHO Expert Committee on Food Additives found the evidence linking aspartame consumption and cancer unconvincing and did not alter intake limits [2]. Although these evaluations were for aspartame and not for other NNS, they called attention more broadly to the continued uncertainty about the role of NNS in general (as a class of sweet-tasting noncaloric replacements for added sugars) in weight management and chronic disease prevention.

Although NNS were once considered inert due to containing no or few calories, NNS have metabolic effects in rodents [3–8] and humans [9,10], and positive associations between NNS consumption and body weight [11,12], metabolic syndrome [13,14], cardiovascular disease [15,16], and stroke [15,17] are reported in observational studies. In some studies, NNS consumption is also positively associated with cancer risk [18,19], although findings of studies linking NNS to cancer outcomes are mixed [20,21]. Several biological mechanisms may contribute to metabolic and health effects of NNS [22,23]. One potential mechanism is through NNS-induced activation of sweet taste receptors, located in the oral cavity as well as throughout the gastrointestinal tract, which stimulates release of hormones involved in appetite regulation and glucose homeostasis [24]. Several experimental studies in humans have shown that consumption of sucralose affects insulin and/or glucose homeostasis [25,26] and may lower insulin sensitivity [9,10]. Another recent randomized controlled trial in humans demonstrated that 2 wk of sucralose or saccharin consumption led to impairments in glucose tolerance, which may be explained by NNS-induced alterations in the gut microbiome [27].

Another proposed mechanism, which has received support in rodent, in vitro, and human studies, is that NNS may promote inflammation [28–30], central to cardiometabolic disease pathophysiology [31,32] and carcinogenesis [33]. For example, incubation of human adipose tissue derived mesenchymal stromal cells with NNS at physiologic concentrations upregulates expression of inflammatory genes [34], promotes adipogenesis [28], and increases reactive oxygen species [28]. In vitro studies also demonstrate that NNS consumption may increase gut permeability and immune reactivity [35], and, several NNS are reported to induce proinflammatory changes in gut bacteria in rodents, which may be modulated by differences in diet composition [36]. Studies in rodents also demonstrate that NNS exposure upregulates lipopolysaccharide biosynthesis genes [30,37,38], a mediator of systemic inflammation, and in humans, NNS consumption dysregulates inflammatory pathways in human adipose tissue [29].

These findings suggest that NNS may contribute to, rather than help to prevent, development of chronic disease. However, research examining NNS consumption in relation to metabolic risk factors and inflammatory biomarkers is limited. The purpose of this study was to examine cross-sectional associations between NNS consumption, metabolic risk factors, and inflammatory biomarkers among adults in the Cancer Prevention Study-3 (CPS-3) Diet Assessment Sub-Study (DAS).

Methods

Study participants

CPS-3 is a prospective cohort study of >300,000 adults aged 30–65 y who were recruited from throughout the United States (35 states plus the District of Columbia and Puerto Rico) between 2006 and 2013 [39]. The CPS-3 DAS was designed to examine the validity and reproducibility of the CPS-3 food frequency questionnaire (FFQ) and studied a subset of 745 CPS-3 participants over 1 y (2015–2016) [40]. DAS participants were recruited from 5 regions of the country (Atlanta, GA; Dallas, TX; Auburn Hills, MI; West Hills, CA; and San Jose, CA) and asked to complete 2 FFQs 1 y apart and 6 24-h interviewer-administered dietary recalls by telephone in the interim; they were also asked to provide 2 fasting (8-h fast) blood samples and 2 24-h urine samples. Blood and urine samples were collected ~6 mo apart, and a 24-h dietary recall was conducted within a week of each fasting blood sample collection, when possible. The CPS-3 DAS protocol was approved by the Institutional Review Board at Emory University (Atlanta, GA, United States), and details of CPS-3 and DAS have been published previously [39,40].

There are a total of 745 participants in the DAS. For the present analysis, participants were excluded for the following: not providing 2 blood samples ($n = 25$); pregnancy ($n = 15$); having diabetes ($n = 43$); taking insulin or another diabetes medication ($n = 8$); or providing incomplete or implausible dietary data ($n = 30$). The incomplete or implausible dietary data exclusion was composed of those who did not complete both FFQs, were missing a full page, 2+ sections, or >100-line items on the FFQ, completed <4 dietary recalls, or reported improbable energy intake on the FFQ (defined as <800 or >4500 kcal for males and <600 or >3800 kcal for females). The final analytic sample included 624 DAS participants.

Assessment of NNS consumption

Twenty-four-hour dietary recalls were administered using the multipass method by trained interviewers at the Pennsylvania State University Dietary Assessment Center and conducted using the Nutrition Data System for Research [41]. Dietary recalls were unannounced and included 4 weekdays and 2 weekend days. The yearlong study period was divided into trimesters; and when possible, 2 24-h dietary recalls were conducted in each trimester to collect data throughout the year and account for seasonal differences in dietary intake and changes in eating behaviors over time. An exception was that 1 dietary recall was conducted within 1 wk of each fasting blood sample collection.

Participants' NNS consumption was estimated using values in Nutrition Data System for Research corresponding to quantities of aspartame, acesulfame-potassium, saccharin, and sucralose in the food and beverage items they reported consuming in each of the dietary recalls collected during the DAS (mean 5.88 recalls per participant). For the main analyses, the mean amounts (mg) of aspartame, acesulfame-potassium, sucralose, and saccharin reported by each participant across the 6 dietary recalls were averaged to estimate total mean daily NNS consumption. Tertiles (low, medium, and high) of total mean daily NNS consumption were then compared with nonconsumption (reported no NNS consumption during any of the dietary recalls). Mean consumption of aspartame, saccharin, and sucralose were also calculated separately for exploratory analyses by type of NNS. Although aspartame cannot be measured in biospecimens due to its rapid degradation following ingestion, metabolites correlated with sucralose, saccharin, and ace-K consumption were previously identified in the DAS [40,42].

Assessment of metabolic and inflammatory outcomes

Blood samples were collected in a fasted state and were refrigerated and transferred to a Quest Diagnostics regional processing laboratory. Blood samples were fractionated by centrifugation, divided into aliquots, and placed in -80^{o}C freezers before being shipped on dry ice to an off-site biorepository for long-term storage in vapor-phase liquid nitrogen.

All biomarkers were measured in the Pollak Research Laboratory at the Jewish General Hospital at McGill University. Circulating concentrations of insulin, C-peptide, leptin, adiponectin, and inflammatory biomarkers, including C-reactive protein (CRP), TNF- α, IL-6, and IL-10, were measured in plasma and hemoglobin A1c (HbA1c) was measured in red blood cells. Insulin was measured using ELISA with reagents from Mercodia AB (Uppsala). The coefficient of variation (CV) for insulin was 3.5%, with intraclass correlations (ICCs) of 98.9%. C-peptide was measured using ELISA with reagents from Ansh Labs LLC (Webster). The CV for C-peptide was 1.6%, with ICCs of 99.4%. Leptin, adiponectin, CRP, TNF- α, and IL-6 were measured using ELISAs with reagents from R&D Systems Inc. (Minneapolis) with CVs of 4.2%, 4.3%, 4.2%, 4.7%, and 4.8%, respectively. Corresponding ICCs were 99.3%, 97.9%, 99.5%, 93.0%, and 95.4%. IL-10 was measured using ELISA with reagents from RayBiotech Life, Inc. (Peachtree Corners). The CV for IL-10 was 4.7%, with ICCs of 97.5%. The Human HbA1c kit (Crystal Chem, Inc.), an enzymatic assay in which lysed whole blood samples are subjected to extensive protease digestion, was used to measure HbA1c. The CV for HbA1c was 2.2%, and the ICC was 82.5%.

Anthropometric assessments

BMI was calculated using participants' self-reported height and weight, which were previously validated [43]. Waist circumference was measured by the participants, in centimeters, using a tape measure they were provided with, along with instructions on how to measure themselves.

Statistical analyses

Descriptive statistics were summarized using means and SDs for continuous variables and using proportions for categorical variables across nonconsumers of NNS and the 3 NNS consumption tertiles. Distributions of all outcomes were examined, and values were log-transformed for insulin, C-peptide, CRP, leptin, adiponectin, TNF-alpha, and IL-6. Mean values of each outcome across the 2 blood samples collected in the DAS were used in all analyses. Multivariable linear regression was used to examine associations between total NNS consumption and all outcomes, continuously and across tertiles of NNS consumption. Base models were adjusted for the following covariates, which were selected a priori: age (y), gender (man or woman), race/ethnicity (American Indian or Alaskan Native; Asian, Native Hawaiian, or Pacific Islander; Black; Hispanic; Non-Hispanic White; or Other), educational attainment (≤high school; some college; completed college; or graduate degree), and physical activity (MET-h per day). Subsequent models were further adjusted for BMI (kg/m^2), diet quality (Healthy Eating Index 2020 [HEI-2020]), and energy intake (kcal/d). Age was calculated from the 2015 survey return data and birthdate, whereas gender, race, and education were self-reported on the enrollment survey. Physical activity was derived from questions on the 2015 survey, and BMI was calculated based on self-reported weight on the 2015 survey and height from the enrollment survey. Diet quality and energy intake were calculated from the 24-h dietary recalls. Age, BMI, diet quality, and energy intake were adjusted as continuous variables, and sex, race, education, smoking status, and physical activity were adjusted as categorical variables. An analogous approach was used to explore associations separately for aspartame and sucralose consumption. Analyses were not conducted separately for saccharin due to only a small number (n = 46) of consumers. Separate analyses were also not performed for ace-K because ace-K is typically consumed in combination with aspartame and/or sucralose. In addition, because NNS use is most common among individuals with obesity, we also stratified analyses by weight status. All statistical analyses were conducted using SAS version 9.4, and associations were considered statistically significant if P values were <0.05.

Results

The mean age of the participants was 52.2 ± 9.6 y, and 63.5% were female, 62.3% were White, and 86% had completed college (Table 1). More than half (55%) reported consuming NNS on at least one of their dietary recalls. Mean NNS consumption was 7, 38, and 221 mg per day among low, medium, and high NNS consumers, respectively.

The predominant type of NNS consumed was aspartame, followed by sucralose and ace-K (Figure 1). Aspartame comprised 80% of all NNS consumption reported in the sample, with sucralose, ace-K, and saccharin accounting for 8.4%, 7.4%, and 4.2%, respectively. Beverages were the main source of NNS in the sample and comprised 61% of reported NNS consumption, followed by food and beverage additions (e.g., NNS packets and NNS-sweetened condiments) (Figure 2). Diet soda was the predominant beverage source of NNS (Supplemental Figure 1), and NNS consumption from food and beverage additions was primarily (88%) in the form of NNS packets added to foods and beverages (e.g., Equal™ and Splenda™). Although foods containing NNS accounted for only 10% of reported NNS consumption, NNS-containing yogurts were the primary food source of NNS, followed by grains and snacks, frozen desserts, nonfrozen desserts, and protein bars (Supplemental Figure 2).

Leptin and CRP concentrations were positively associated with NNS consumption prior to adjustment for BMI, HEI-2020, and energy intake (P-trend = 0.0006 and P-trend = 0.02 for leptin and CRP, respectively), but associations were attenuated in fully adjusted models (Table 2). NNS consumption was not associated with insulin, C-peptide, HbA1c, adiponectin, TNF-α, IL-6, or IL-10 in base or fully adjusted models. In analyses stratified by BMI (Supplemental Table 1), NNS consumption was positively associated with IL-6 among participants with BMI ≥25 kg/m^2 but not among participants with BMI <25 kg/m^2.

TABLE 1
Characteristics of participants (n = 624) in the Cancer Prevention Study-3 Diet Assessment Substudy, by nonnutritive sweetener (NNS) consumption.

Characteristics[1]	Overall sample (n = 624)	Nonconsumer (n = 283)	Low consumers <15.5 mg NNS per day (n = 113)	Medium consumers 15.5 mg – <69.0 mg NNS per day (n = 114)	High consumers ≥69.0 mg NNS per day (n = 114)
Age, y	52.2 ± 9.6	52.6 ± 9.5	51.3 ± 10.0	51.5 ± 9.6	52.9 ± 9.6
Sex					
Male	228 (36.5)	110 (48.2)	40 (17.5)	31 (13.6)	47 (20.6)
Female	396 (63.5)	173 (43.7)	73 (18.4)	83 (21.0)	67 (16.9)
Race/ethnicity					
White	389 (62.3)	166 (42.7)	63 (16.2)	78 (20.1)	82 (21.1)
Black	143 (22.9)	83 (58.0)	26 (18.2)	18 (12.6)	16 (11.2)
Hispanic	92 (14.7)	34 (37.0)	24 (26.1)	18 (19.6)	16 (17.4)
Education					
< College	87 (14.0)	37 (42.5)	14 (16.1)	21 (24.1)	15 (17.2)
Completed college	271 (43.5)	127 (46.9)	47 (17.3)	54 (19.9)	43 (15.9)
Graduate degree	265 (42.5)	118 (44.5)	52 (19.6)	39 (14.7)	56 (21.1)
Smoking status					
Never	464 (74.6)	207 (44.6)	89 (19.2)	81 (17.5)	87 (18.8)
Current	22 (3.5)	12 (54.5)	2 (9.1)	2 (9.1)	6 (27.3)
Former	136 (21.9)	64 (47.1)	21 (15.4)	31 (22.8)	20 (14.7)
Physical activity, MET-h/wk					
< 5	101 (16.3)	44 (43.6)	24 (23.8)	14 (13.9)	19 (18.8)
5 – <10	12 (1.9)	7 (58.3)	2 (16.7)	0 (0)	3 (25)
10 – <15	69 (11.1)	31 (44.9)	9 (13)	14 (10.3)	15 (21.7)
≥15	439 (70.7)	200 (45.6)	78 (17.8)	85 (19.4)	76 (17.3)
Energy intake[2], kcal/d	1896 ± 509	1938 ± 497	1833 ± 515	1845 ± 483	1909 ± 554
Total NNS[2], mg/d	48.5 ± 106.8	0.0 ± 0.0	6.9 ± 4.5	37.7 ± 15.1	220.7 ± 157.9
Aspartame[2], mg/d	38.8 ± 99.0	0.0 ± 0.0	2.4 ± 4.1	24.2 ± 18.5	185.8 ± 163.0
Sucralose[2], mg/d	4.0 ± 10.1	0.0 ± 0.0	3.0 ± 3.9	8.4 ± 12.5	10.6 ± 16.9
Saccharin[2], mg/d	2.0 ± 11.4	0.0 ± 0.0	0.6 ± 2.5	1.3 ± 6.4	9.2 ± 24.7
Ace-K[2], mg/d	3.6 ± 11.0	0.0 ± 0.0	0.8 ± 1.4	3.8 ± 6.4	15.0 ± 21.3
Added sugar[2], g/d	44.0 ± 24.4	44.8 ± 25.5	43.0 ± 25.5	43.3 ± 20.7	43.6 ± 24.3
HEI-2020 score (max: 100)	66.4 ± 12.9	67.1 ± 13.8	66.9 ± 12.6	65.6 ± 13.0	64.9 ± 10.6
BMI, kg/m^2	27.3 ± 5.8	26.9 ±6.0	27.5 ± 5.0	27.0 ± 5.2	28.0 ±6.4
Waist circumference, cm	93.2 ± 15.3	92.8 ±16.1	92.8 ± 12.8	92.3 ± 14.1	95.4 ± 16.4

Daily mean NNS consumption of ~7 mg, 38 mg, and 221 mg among low, medium, and high NNS consumers, respectively.
Abbreviations: Ace-K, Acesulfame-potassium; ACS, American Cancer Society; NNS, nonnutritive sweetener; MET-h, metabolic equivalent hour; SSB, sugar-sweetened beverage.
[1] Values are the mean ± SD for continuous variables, and frequency (%) for categorical variables.
[2] Average value from 24-h dietary recalls conducted throughout the DAS. The average value was used for waist circumference, which was measured twice 6 months apart.

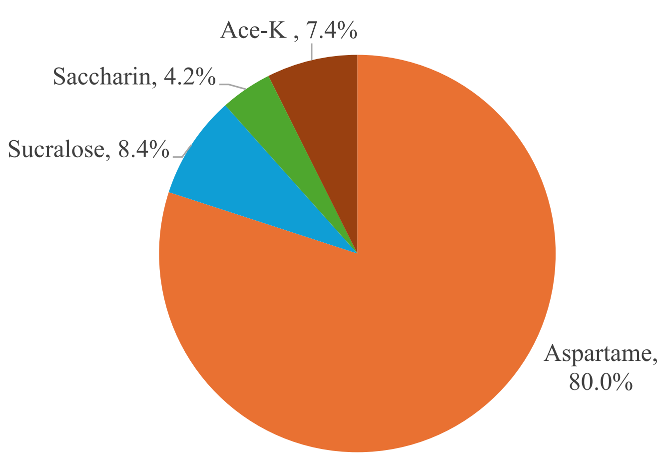

Figure 1. Aspartame in orange, sucralose in blue, ace-K in red, and saccharin in green.

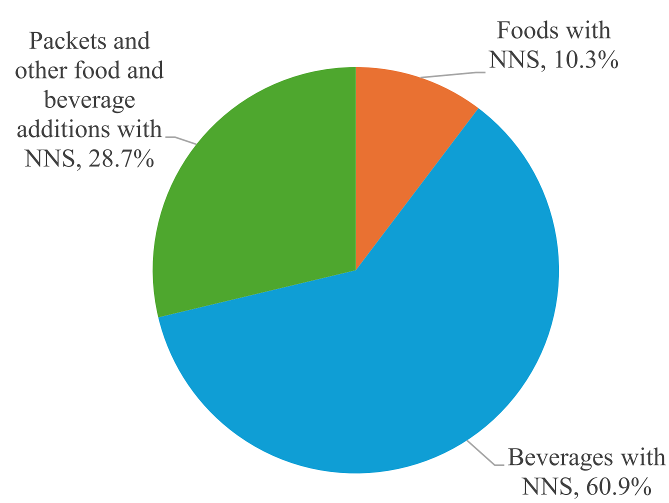

Figure 2. Beverages in blue, packets in green, and foods in red.

In exploratory sweetener-specific analyses examining associations by aspartame consumption (Table 3), results were similar to those for total NNS consumption. Leptin was positively associated with aspartame consumption before (P-trend = 0.0004) adjustment for BMI, HEI-2020, and energy intake, but associations were attenuated in fully adjusted models. No associations between aspartame consumption and insulin, C-

TABLE 2
Non-nutritive sweetener consumption in relation to cardiometabolic risk factors, overall and across tertiles of consumption.

	Nonconsumers (mean, 95% CI) $n = 283$	Low consumers (mean, 95% CI) $n = 113$		Medium consumers (mean, 95% CI) $n = 114$		High consumers (mean, 95% CI) $n = 114$		*P*-trend	β ± SE[3]	*P* value
		Estimate (95% CI)	*P* value	Estimate (95% CI)	*P* value	Estimate (95% CI)	*P* value			
Insulin[1],μU/mL	1.8 (1.5, 2.2)	1.8 (1.4, 2.2)	0.43	1.9 (1.5, 2.3)	0.47	1.9 (1.5, 2.3)	0.41	0.32	0.002 ± 0.004	0.67
Insulin[2],μU/mL	1.8 (1.5, 2.1)	1.7 (1.4, 2.0)	0.09	1.8 (1.5, 2.1)	0.95	1.7 (1.4, 2.0)	0.26	0.41	−0.004 ± 0.004	0.23
C-Peptide[1], mg/dL	1.1 (0.8, 1.4)	1.0 (0.8, 1.3)	0.39	1.1 (0.8, 1.3)	0.78	1.1 (0.8, 1.4)	0.73	0.83	0.001 ± 0.003	0.80
C-Peptide[2], mg/dL	1.0 (0.8, 1.3)	1.0 (0.7, 1.2)	0.08	1.0 (0.7, 1.2)	0.28	1.0 (0.7, 1.2)	0.14	0.12	−0.003 ± 0.003	0.23
C-reactive protein[1], ng/dL	7.3 (6.52, 8.14)	7.5 (6.7, 8.3)	0.24	7.5 (6.7, 8.3)	0.24	7.7 (6.8, 8.5)	0.02	0.02	0.02 ± 0.01	0.10
C-reactive protein[2], ng/dL	7.2 (6.5. 7.9)	7.3 (6.5, 8.0)	0.51	7.3 (6.5, 8.0)	0.49	7.3 (6.6, 8.1)	0.37	0.33	0.003 ± 0.008	0.68
HbA1C[1],%	5.6 (5.2, 6.0)	5.5 (5.1, 5.9)	0.12	5.5 (5.1, 5.9)	0.21	5.6 (5.2, 6.0)	0.93	0.74	0.004 ± 0.005	0.35
HbA1C[2],%	5.6 (5.2, 6.0)	5.5 (5.1, 5.9)	0.10	5.5 (5.1, 5.9)	0.17	5.6 (5.1, 6.0)	0.70	0.44	0.002 ± 0.005	0.62
Leptin[1], ng/mL	9.4 (8.9, 10.0)	9.7 (9.1, 10.2)	0.02	9.6 (9.0, 10.2)	0.09	9.8 (9.2, 10.3)	0.0005	<0.01	0.02 ± 0.01	0.01
Leptin[2], ng/mL	9.3 (8.8, 9.7)	9.3 (8.9, 9.7)	0.23	9.3 (8.9, 9.7)	0.47	9.3 (8.9, 9.7)	0.47	0.41	0.003 ± 0.005	0.56
Adiponectin[1],μg/mL	8.7 (8.4, 9.0)	8.8 (8.4, 9.1)	0.11	8.7 (8.3, 9.0)	0.86	8.7 (8.4, 9.0)	0.99	0.90	−0.002 ± 0.004	0.58
Adiponectin[2],μg/mL	8.7 (8.4, 9.0)	8.8 (8.5, 9.1)	0.04	8.7 (8.4, 9.0)	0.85	8.7 (8.5, 9.1)	0.31	0.42	0.0009 ± 0.003	0.79
TNF-α[1], pg/mL	0.3 (0.1, 0.5)	0.3 (0.1, 0.5)	0.37	0.3 (0.1, 0.5)	0.48	0.3 (0.1, 0.5)	0.25	0.23	0.002 ± 0.002	0.27
TNF-α[2], pg/mL	0.3 (0.1, 0.4)	0.3 (0.1, 0.5)	0.64	0.3 (0.1, 0.5)	0.73	0.3 (0.1, 0.5)	0.85	0.79	0.0008 ± 0.002	0.71
IL-6[1], pg/mL	0.5 (0.1, 0.8)	0.5 (0.1, 0.9)	0.67	0.5 (0.1, 0.9)	0.40	0.6 (0.2, 0.9)	0.09	0.09	0.007 ± 0.004	0.09
IL-6[2], pg/mL	0.4 (0.1, 0.8)	0.5 (0.1, 0.8)	0.93	0.5 (0.1, 0.9)	0.50	0.5 (0.1, 0.8)	0.49	0.41	0.004 ± 0.004	0.38
IL-10[1], pg/mL	0.3 (0.2, 0.3)	0.3 (0.2, 0.3)	0.55	0.3 (0.2, 0.3)	0.57	0.3 (0.2, 0.3)	0.74	0.94	0.0002 ± 0.0007	0.79
IL-10[2], pg/mL	0.3 (0.2, 0.3)	0.3 (0.2, 0.3)	0.46	0.3 (0.2, 0.3)	0.54	0.3 (0.2, 0.3)	0.77	0.97	0.0002 ± 0.0007	0.73

Total daily NNS consumption of ~7mg, 38mg, and 221mg among low, medium, and high NNS consumers, respectively. NNS consumption was < 15.5 mg, 15.5–69.0 mg, and ≥ 69.0 mg among low, medium and high NNS consumers, respectively. Values were log-transformed for insulin, C-peptide, C-reactive protein, leptin, adiponectin, TNF-alpha, and IL-6.
Abbreviations: CI, confidence interval; IL, interleukin.
[1] Adjusted for age, sex, race, education, smoking, and physical activity.
[2] Adjusted for age, sex, race, education, smoking, physical activity, diet quality, energy intake, and BMI.
[3] Per 20 mg/d total NNS.

TABLE 3
Aspartame consumption in relation to cardiometabolic risk factors, overall and across tertiles of aspartame consumption.

	Non-NNS consumers (mean, 95% CI) $n = 366$	Low aspartame consumers, <19.5 mg per day (mean, 95% CI) $n = 86$		Medium aspartame consumers, 19.5–77.7 my per day (mean, 95% CI) $n = 86$		High aspartame consumers, ≥77.7 mg per day (mean, 95% CI) $n = 86$		*P*-trend	β ± SE[3]	*P* value
		Estimate (95% CI)	*P* value	Estimate (95% CI)	*P* value	Estimate (95% CI)	*P* value			
Insulin[1],μU/mL	1.8 (1.5, 2.20)	1. 8 (1.4, 2.2)	0.35	1.90 (1.53, 2.28)	0.35	1.92 (1.54, 2.30)	0.29	0.26	0.002 ± 0.005	0.63
Insulin[2],μU/mL	1.8 (1.5, 2.1)	1.7 (1.3, 2.0)	0.12	1.8 (1.5, 2.1)	0.70	1.7 (1.4, 2.1)	0.62	0.73	−0.003 ± 0.004	0.44
C-Peptide[1], mg/dL	1.1 (0.80, 1.3)	1.04 (0.8, 1.3)	0.57	1.08 (0.8, 1.4)	0.92	1.12 (0.8, 1.4)	0.43	0.55	0.002 ± 0.004	0.64
C-Peptide[2], mg/dL	1.0 (0.8, 1.3)	1.0 (0.7, 1.2)	0.27	1.0 (0.7, 1.2)	0.59	1.0 (0.7, 1.2)	0.56	0.44	−0.002 ± 0.003	0.56
C-reactive protein[1], ng/dL	7.4 (6.6, 8.2)	7.4 (6.5, 8.2)	0.90	7.5 (6.7, 8.4)	0.34	7.5 (6.7, 8.4)	0.33	0.26	0.008 ± 0.01	0.42
C-reactive protein[2], ng/dL	7.2 (6.5, 8.0)	7.2 (6.4, 7.9)	0.60	7.3 (6.5, 8.0)	0.63	7.2 (6.4, 7.9)	0.73	0.91	−0.002 ± 0.009	0.84
HbA1C[1],%	5.6 (5.2, 6.0)	5.4 (5.0, 5.9)	0.03	5.6 (5.1, 6.0)	0.57	5.6 (5.2, 6.0)	0.94	0.72	0.006 ± 0.005	0.30
HbA1C[2],%	5.6 (5.2, 6.0)	5.4 (5.0, 5.9)	0.03	5.5 (5.1, 6.0)	0.49	5.6 (5.1, 6.0)	0.70	0.44	0.003 ± 0.005	0.54
Leptin[1], ng/mL	9.5 (8.9, 10.0)	9. 7 (9.1, 10.3)	0.06	9.7 (9.1, 10.2)	0.06	9.9 (9.3, 10.4)	0.001	<0.01	0.01 ± 0.01	0.04
Leptin[2], ng/mL	9.3 (8. 9, 9.7)	9.3 (8.8, 9.6)	0.67	9.2 (8.8, 9.6)	0.15	9.2 (8.8, 9.6)	0.29	0.12	0.05 ± 0.05	0.34
Adiponectin[1],μg/mL	8.7 (8.4, 9.0)	8.8 (8.4, 9.1)	0.26	8.6 (8.3, 9.0)	0.45	8.7 (8.4, 9.0)	0.85	0.71	−0.002 ± 0.004	0.60
Adiponectin[2],μg/mL	8.7 (8.4, 9.0)	8.8 (8.5, 9.1)	0.43	8.8 (8.4, 9.1)	0.59	8.7 (8.4, 9.0)	0.91	0.67	0.03 ± 0.04	0.44
TNF-α[1], pg/mL	0.3 (0.1, 0.5)	0.3 (0.1, 0.5)	0.52	0.3 (0.1, 0.5)	0.94	0.3 (0.1, 0.5)	0.70	0.57	0.002 ± 0.003	0.38
TNF-α[2], pg/mL	0.3 (0.1, 0.5)	0.3 (0.1, 0.5)	0.90	0.3 (0.1, 0.5)	0.77	0.3 (0.1, 0.5)	0.90	0.84	0.0008 ± 0.002	0.74
IL-6[1], pg/mL	0.5 (0.1, 0.9)	0. (0.1, 0.9)	0.72	0.5 (0.1, 0.9)	0.93	0.5 (0.1, 0.9)	0.58	0.68	0.004 ± 0.005	0.41
IL-6[2], pg/mL	0.4 (0.1, 0.8)	0.5 (0.1, 0.8)	0.77	0.4 (0.1, 0.8)	0.67	0.4 (0.1, 0.8)	0.73	0.67	0.0004 ± 0.005	0.93
IL-10[1], pg/mL	0.3 (0.2, 0.3)	0.2 (0.2, 0.3)	0.04	0.3 (0.2, 0.3)	0.84	0.3 (0.2, 0.3)	0.63	0.54	−0.00002 ± 0.0008	0.98
IL-10[2], pg/mL	0.3 (0.2, 0.3)	0.2 (0.2, 0.3)	0.04	0.3 (0.2, 0.3)	0.81	0.3 (0.2, 0.3)	0.67	0.54	0.00005 ± 0.0008	0.94

Mean daily aspartame consumption of ~8mg, 44mg, and 230mg among low, medium, and high NNS consumers, respectively. Values were log-transformed for insulin, C-peptide, C-reactive protein, leptin, adiponectin, TNF-alpha, and IL-6.
Abbreviation: IL, interleukin; TNF, tumor necrosis factor.
1 Adjusted for age, sex, race, education, smoking, and physical activity.
2 Adjusted for age, sex, race, education, smoking, physical activity, BMI, diet quality, and energy intake.
3 Per 20 mg/d aspartame.

peptide, HbA1c, adiponectin, CRP, TNF-α, IL-6, or IL-10 were observed.

In exploratory analyses examining associations across sucralose consumption (Table 4), sucralose consumption was inversely associated with insulin (P-trend = 0.01) and C-peptide (P-trend = 0.01) concentrations in fully adjusted models. However, this relationship appeared to be driven by higher values among nonconsumers, and there was no apparent trend among consumers.

Discussion

These cross-sectional findings demonstrate that while NNS consumption is positively associated with leptin and CRP prior to adjustment for BMI, diet quality, and energy intake, the findings were overall null. However, in participants with overweight or obesity (i.e., BMI $\geq$ 25 kg/m^2), NNS consumption was positively associated with IL-6. This suggests that individuals with excess adiposity or pre-existing metabolic perturbations may be more sensitive to unfavorable effects of NNS on metabolic and inflammatory biomarkers. Indeed, differential effects of NNS consumption on metabolic outcomes in individuals with obesity compared with normal weight have been reported previously [44,45].

The positive association between NNS consumption and CRP prior to adjustment for BMI, diet quality, and energy intake are notable considering the findings of mechanistic studies showing that various NNS may enrich the expression of proinflammatory cytokines and exacerbate systemic inflammation [30,46]. Potential proinflammatory effects of NNS are proposed to be mediated by NNS-induced changes in gut microbiota [30, 47–49].

A surprising finding was that sucralose consumption was inversely associated with insulin and C-peptide and was not associated with HbA1c, which suggests a potential benefit for insulin sensitivity. This is in contrast to several prior studies demonstrating that consumption of sucralose augments glucose-stimulated insulin responses [26,50,51], impairs glucose tolerance [26,27,50], and reduces insulin sensitivity [9,10,52]. Explanations for lower insulin and C-peptide concentrations observed with higher sucralose consumption in the present analysis are unclear and warrant further investigation; as the relationship appeared nonlinear, these findings may be due to chance.

Given that aspartame comprised the vast majority of NNS consumption among DAS participants, findings of analyses across tertiles of aspartame consumption mostly mirrored those observed for overall NNS consumption. However, in contrast to the findings for overall NNS consumption, no associations between aspartame consumption and inflammatory biomarkers (i.e., CRP) were observed. This is noteworthy because most studies demonstrating potential proinflammatory effects of NNS have administered sucralose, saccharin, and/or acesulfame-potassium [30,46,53], and underscores the likelihood that different types of NNS may have variable implications for metabolic health [54]. Although findings of analyses by sweetener type are thought-provoking and emphasize the need to differentiate between types of NNS when investigating their metabolic and health effects, associations should be interpreted cautiously due to the small sample size. Furthermore, many NNS consumers ingest multiple types of NNS, making it difficult, if not impossible, to reliably disentangle their effects in observational studies [54].

Several additional considerations may also explain the null findings and/or relatively weak associations linking overall NNS consumption to metabolic outcomes in the main analysis. First, reported NNS consumption was relatively modest, with mean daily intakes across low, medium, and high tertiles of NNS consumption of ∼7 mg/d (equivalent to <1 NNS-containing sweetener packet), 38 mg/d (∼1/2 to 1/3 of a diet beverage, depending on the type of NNS), and 221 mg/d (∼2 to 3 diet sodas, depending on the type of NNS), respectively. In addition, individuals with diabetes, among whom NNS consumption is prevalent, were excluded from the analyses. It is also notable that reported NNS consumption was from a variety of sources, yet most consumption was driven by diet sodas, the majority of which are sweetened with aspartame. Furthermore, as described above, some individuals, particularly those with pre-existing obesity and/or metabolic perturbations, may be more susceptible to adverse effects of NNS consumption [44, 45], and future research to understand heterogeneity in NNS effects across population subgroups is warranted.

The present analyses have several strengths, including the robust dietary data collected using repeated dietary recalls over a yearlong period in a racially/ethnically and geographically diverse sample, which allowed us to capture a variety of dietary sources of NNS and provided the opportunity to conduct exploratory analyses across different types of NNS. Using metabolic and inflammatory biomarker concentrations averaged across 2 blood samples also enhanced the reliability of biomarker values and examining associations both continuously and across tertiles of NNS consumption strengthened the analysis.

Key limitations include the inability to assess other relevant metabolic outcomes such as glucose, insulin resistance, and lipid profiles. In addition, we were unable to compare results across different sources of NNS because most participants who consumed NNS-containing foods also consumed NNS-containing beverages and/or packets. The sample size of the DAS was also relatively small compared with other cohorts.

Taken together, associations between NNS consumption and leptin and CRP were attenuated after adjustment for BMI, diet quality, and energy intake in the full sample, and no associations with insulin, C-peptide, HbA1c, TNF-α, adiponectin, IL-6, or IL-10 were observed. However, in participants who had overweight or obesity, NNS consumption was positively associated with leptin and IL-6, and associations with IL-6 remained statistically significant in fully adjusted models. Given the cross-sectional nature of these analyses, further investigations assessing NNS consumption and changes in metabolic and inflammatory biomarkers over time are needed, ideally by type of NNS.

TABLE 4
Sucralose consumption in relation to cardiometabolic risk factors, overall and across tertiles of sucralose consumption.

	Non-NNS consumers (mean, 95% CI) $n = 436$	Low sucralose consumers, <4.5 mg per day (mean, 95% CI) $n = 63$		Medium sucralose consumers, 4.5–14.1 mg per day (mean, 95% CI) $n = 65$		High sucralose consumers, ≥14.1 mg per day (mean, 95% CI) $n = 60$		*P* trend	β ± SE[3]	*P* value
		Estimate (95% CI)	*P* value	Estimate (95% CI)	*P* value	Estimate (95% CI)	*P* value			
Insulin[1], μU/mL	1.9 (1.5, 2.2)	1.8 (1.4, 2.2)	0.18	1.9 (1.5, 2.3)	0.86	1.8 (1.4, 2.1)	0.13	0.15	−0.04 ± 0.04	0.34
Insulin[2], μU/mL	1.8 (1.5, 2.1)	1. 7 (1.3, 2.0)	0.07	1. 7 (1.3, 2.0)	0.08	1.7 (1.3, 2.0)	0.07	0.01	−0.06 ± 0.04	0.10
C-Peptide[1], mg/dL	1.1 (0.8, 1.4)	1.0 (0.7, 1.3)	0.21	1.1 (0.8, 1.4)	0.44	1.0 (0.7, 1.3)	0.14	0.10	−0.02 ± 0.03	0.55
C-Peptide[2], mg/dL	1.0 (0.8, 1.3)	0.9 (0.7, 1.2)	0.11	0.9 (0.7, 1.2)	0.04	0.9 (0.7, 1.2)	0.10	0.02	−0.03±0.03	0.30
C-reactive protein[1], ng/dL	7.4 (6.6, 8.2)	7.5 (6.6, 8.3)	0.89	7.6 (6.8, 8.5)	0.26	7.4 (6.6, 8.3)	0.95	0.56	0.06 ± 0.10	0.56
C-reactive protein[2], ng/dL	7.2 (6.5, 8.0)	7.2 (6.5, 8.0)	0.92	7.2 (6.5, 8.0)	0.98	7.2 (6.5, 8.0)	0.98	0.99	0.02 ± 0.09	0.83
HbA1C[1],%	5.6 (5.2, 6.0)	5.6 (5.1, 6.0)	0.83	5.5 (5.1, 6.0)	0.49	5.6 (5.1, 6.0)	0.83	0.61	−0.01 ± 0.05	0.84
HbA1C[2],%	5.6 (5.2, 6.0)	5.6 (5.1, 6.0)	0.81	5.5 (5.0, 5.9)	0.23	5.6 (5.1, 6.0)	0.76	0.41	−0.02 ± 0.05	0.64
Leptin[1], ng/mL	9.6 (9.0, 10.1)	9.6 (9.0, 10.2)	0.81	9.7 (9.1, 10.3)	0.32	9.5 (8.9, 10.1)	0.39	0.86	−0.01 ± 0.07	0.84
Leptin[2], ng/mL	9.3 (8.9, 9.7)	9.3 (8.8, 9.6)	0.67	9.2 (8.8, 9.6)	0.15	9.2 (8.8, 9.6)	0.29	0.12	−0.05 ± 0.05	0.34
Adiponectin[1],μg/mL	8.7 (8.4, 9.0)	8.7 (8.4, 9.0)	0.49	8.7 (8.4, 9.0)	0.93	8.7 (8.4, 9.0)	0.92	0.90	0.02 ± 0.04	0.60
Adiponectin[2],μg/mL	8.7 (8.4, 9.0)	8.8 (8.5, 9.1)	0.43	8.8 (8.4, 9.1)	0.59	8.7 (8.4, 9.0)	0.91	0.67	0.03 ± 0.04	0.44
TNF-α[1], pg/mL	0.3 (0.1, 0.5)	0.3 (0.1, 0.5)	0.87	0.3 (0.1, 0.5)	0.32	0.3 (0.1, 0.5)	0.82	0.51	0.02 ± 0.02	0.36
TNF-α[2], pg/mL	0.3 (0.1, 0.5)	0.3 (0.1, 0.5)	0.98	0.3 (0.1, 0.5)	0.73	0.3 (0.1, 0.5)	0.80	0.73	0.02 ± 0.02	0.44
IL-6[1], pg/mL	0.5 (0.1, 0.8)	0.5 (0.1, 0.9)	0.50	0.5 (0.1, 0.9)	0.37	0.6 (0.2, 1.0)	0.08	0.06	0.09 ± 0.05	0.06
IL-6[2], pg/mL	0.4 (0.1, 0.8)	0.5 (0.1, 0.8)	0.62	0.4 (0.0, 0.8)	0.92	0.6 (0.2, 0.9)	0.08	0.16	0.07 ± 0.04	0.11
IL-10[1], pg/mL	0.3 (0.2, 0.3)	0.3 (0.2, 0.3)	0.10	0.3 (0.2, 0.3)	0.49	0.3 (0.2, 0.3)	0.88	0.53	−0.01 ± 0.01	0.27
IL-10[2], pg/mL	0.3 (0.2, 0.3)	0.3 (0.2, 0.3)	0.11	0.3 (0.2, 0.3)	0.43	0.3 (0.2, 0.3)	0.83	0.47	−0.007 ± 0.007	0.33

Mean daily sucralose consumption of ~2mg, 9mg, and 30mg among low, medium, and high NNS consumers, respectively. Values were log-transformed for insulin, C-peptide, C-reactive protein, leptin, adiponectin, TNF-alpha, and IL-6.
Abbreviation: IL, interleukin; TNF, tumor necrosis factor.
1 Adjusted for age, sex, race, education, smoking, and physical activity.
2 Adjusted for age, sex, race, education, smoking, physical activity, BMI, diet quality, and energy intake.
3 Per 20 mg/d sucralose.

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Author Contributions

The authors' responsibilities were as follows – ACS, YW, CYU, RAH, CL, MLM: designed the research; AR, DM, AN, YW, MP: analyzed the data; ACS: wrote the first draft of the manuscript; The authors assume full responsibility for all analyses and interpretation of results; and all authors: have read and approved the final manuscript.

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Conflicts of interest

AS reports financial support was provided by American Cancer Society. All other authors report no conflicts of interest.

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Data availability

Data are available from the American Cancer Society by following the ACS Data Access Procedures (https://www.cancer.org/content/dam/cancer-org/research/epidemiology/cancer-prevention-study-data-access-policies.pdf) for researchers who meet the criteria for access to confidential data. Please email cohort.data@cancer.org to inquire about access.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tjnut.2025.03.020.

References

[1] World Health Organization, Use of non-sugar sweeteners: WHO guideline, World Health Organization. [Internet], Geneva, 2023 [Last accessed: April 24th, 2025]. Available from: https://www.who.int/publications/i/item/9789240073616.

[2] World Health Organization, Aspartame hazard and risk assessment results released, [Internet]. [cited Apirl 24th, 2025]. Available from: https://www.who.int/news/item/14-07-2023-aspartame-hazard-and-risk-assessment-results-released.

[3] S.E. Swithers, A.F. Laboy, K. Clark, S. Cooper, T.L. Davidson, Experience with the high-intensity sweetener saccharin impairs glucose homeostasis and GLP-1 release in rats, Behav. Brain Res. 233 (1) (2012) 1–14, https://doi.org/10.1016/j.bbr.2012.04.024.

[4] S.E. Swithers, Artificial sweeteners produce the counterintuitive effect of inducing metabolic derangements, Trends Endocrinol. Metab. 24 (9) (2013) 431–441, https://doi.org/10.1016/j.tem.2013.05.005.

[5] J. Suez, T. Korem, D. Zeevi, G. Zilberman-Schapira, C.A. Thaiss, O. Maza, et al., Artificial sweeteners induce glucose intolerance by altering the gut microbiota, Nature 514 (7521) (2014) 181–186, https://doi.org/10.1038/nature13793.

[6] K. Mitsutomi, T. Masaki, T. Shimasaki, K. Gotoh, S. Chiba, T. Kakuma, et al., Effects of a nonnutritive sweetener on body adiposity and energy metabolism in mice with diet-induced obesity, Metabolism 63 (1) (2014) 69–78, https://doi.org/10.1016/j.metabol.2013.09.002.

[7] M.S. Palmnas, T.E. Cowan, M.R. Bomhof, J. Su, R.A. Reimer, H.J. Vogel, et al., Low-dose aspartame consumption differentially affects gut microbiota-host metabolic interactions in the diet-induced obese rat, PLoS One 9 (10) (2014) e109841, https://doi.org/10.1371/journal.pone.0109841.

[8] W.N. Cong, R. Wang, H. Cai, C.M. Daimon, M. Scheibye-Knudsen, V.A. Bohr, et al., Long-term artificial sweetener acesulfame potassium treatment alters neurometabolic functions in C57BL/6J mice, PLoS One 8 (8) (2013) e70257, https://doi.org/10.1371/journal.pone.0070257.

[9] A. Romo-Romo, C.A. Aguilar-Salinas, G.X. Brito-Cordova, R.A. Gomez-Diaz, P. Almeda-Valdes, Sucralose decreases insulin sensitivity in healthy subjects: a randomized controlled trial, Am. J. Clin. Nutr. 108 (3) (2018) 485–491, https://doi.org/10.1093/ajcn/nqy152.

[10] A. Lertrit, S. Srimachai, S. Saetung, S. Chanprasertyothin, L.O. Chailurkit, C. Areevut, et al., Effects of sucralose on insulin and glucagon-like peptide-1 secretion in healthy subjects: a randomized, double-blind, placebo-controlled trial, Nutrition 55-56 (2018) 125–130, https://doi.org/10.1016/j.nut.2018.04.001.

[11] S.P. Fowler, K. Williams, R.G. Resendez, K.J. Hunt, H.P. Hazuda, M.P. Stern, Fueling the obesity epidemic? Artificially sweetened beverage use and long-term weight gain, Obesity (Silver Spring) 16 (8) (2008) 1894–1900, https://doi.org/10.1038/oby.2008.284.

[12] R.A. Forshee, M.L. Storey, Total beverage consumption and beverage choices among children and adolescents, Int. J. Food Sci. Nutr. 54 (4) (2003) 297–307, https://doi.org/10.1080/09637480120092143.

[13] P.L. Lutsey, L.M. Steffen, J. Stevens, Dietary intake and the development of the metabolic syndrome: the atherosclerosis risk in communities study, Circulation 117 (6) (2008) 754–761, https://doi.org/10.1161/CIRCULATIONAHA.107.716159.

[14] J.A. Nettleton, P.L. Lutsey, Y. Wang, J.A. Lima, E.D. Michos, D.R. Jacobs Jr., Diet soda intake and risk of incident metabolic syndrome and type 2 diabetes in the Multi-Ethnic Study of Atherosclerosis (MESA), Diabetes Care 32 (4) (2009) 688–694, https://doi.org/10.2337/dc08-1799.

[15] H. Gardener, T. Rundek, M. Markert, C.B. Wright, M.S. Elkind, R.L. Sacco, Diet soft drink consumption is associated with an increased risk of vascular events in the Northern Manhattan study, J. Gen. Intern. Med. 27 (9) (2012) 1120–1126, https://doi.org/10.1007/s11606-011-1968-2.

[16] J. Yin, Y. Zhu, V. Malik, X. Li, X. Peng, F.F. Zhang, et al., Intake of sugar-sweetened and low-calorie sweetened beverages and risk of cardiovascular disease: a meta-analysis and systematic review, Adv. Nutr. 12 (1) (2021) 89–101, https://doi.org/10.1093/advances/nmaa084.

[17] M.P. Pase, J.J. Himali, A.S. Beiser, H.J. Aparicio, C.L. Satizabal, R.S. Vasan, et al., Sugar- and artificially sweetened beverages and the risks of incident stroke and dementia: a prospective cohort study, Stroke 48 (5) (2017) 1139–1146, https://doi.org/10.1161/STROKEAHA.116.016027.

[18] C. Debras, E. Chazelas, B. Srour, N. Druesne-Pecollo, Y. Esseddik, F. Szabo de Edelenyi, et al., Artificial sweeteners and cancer risk: results from the NutriNet-Sante population-based cohort study, PLoS Med 19 (3) (2022) e1003950, https://doi.org/10.1371/journal.pmed.1003950.

[19] J.K. Bassett, R.L. Milne, D.R. English, G.G. Giles, A.M. Hodge, Consumption of sugar-sweetened and artificially sweetened soft drinks and risk of cancers not related to obesity, Int. J. Cancer 146 (12) (2020) 3329–3334, https://doi.org/10.1002/ijc.32772.

[20] A.M. Hodge, J.K. Bassett, R.L. Milne, D.R. English, G.G. Giles, Consumption of sugar-sweetened and artificially sweetened soft drinks and risk of obesity-related cancers, Public Health Nutr 21 (9) (2018) 1618–1626, https://doi.org/10.1017/S1368980017002555.

[21] E.M. Navarrete-Munoz, P.A. Wark, D. Romaguera, N. Bhoo-Pathy, D. Michaud, E. Molina-Montes, et al., Sweet-beverage consumption and risk of pancreatic cancer in the European Prospective Investigation into Cancer and Nutrition (EPIC), Am. J. Clin. Nutr. 104 (3) (2016) 760–768, https://doi.org/10.3945/ajcn.116.130963.

[22] M.Y. Pepino, Metabolic effects of non-nutritive sweeteners, Physiol. Behav. 152 (2015) 450–455, https://doi.org/10.1016/j.physbeh.2015.06.024.

[23] K.I. Rother, E.M. Conway, A.C. Sylvetsky, How non-nutritive sweeteners influence hormones and health, Trends Endocrinol. Metab. 29 (7) (2018) 455–467, https://doi.org/10.1016/j.tem.2018.04.010.

[24] I. Kojima, Y. Nakagawa, The role of the sweet taste receptor in enteroendocrine cells and pancreatic beta-cells, Diabetes Metab. J 35 (5) (2011) 451–457, https://doi.org/10.4093/dmj.2011.35.5.451.

[25] N. Bueno-Hernandez, M. Esquivel-Velazquez, R. Alcantara-Suarez, A.Y. Gomez-Arauz, A.J. Espinosa-Flores, K.L. de Leon-Barrera, et al., Chronic sucralose consumption induces elevation of serum insulin in young healthy adults: a randomized, double blind, controlled trial,

Nutr. J. 19 (1) (2020) 32, https://doi.org/10.1186/s12937-020-00549-5.
[26] M.Y. Pepino, C.D. Tiemann, B.W. Patterson, B.M. Wice, S. Klein, Sucralose affects glycemic and hormonal responses to an oral glucose load, Diabetes Care 36 (9) (2013) 2530–2535, https://doi.org/10.2337/dc12-2221.
[27] J. Suez, Y. Cohen, R. Valdes-Mas, U. Mor, M. Dori-Bachash, S. Federici, et al., Personalized microbiome-driven effects of non-nutritive sweeteners on human glucose tolerance, Cell 185 (18) (2022) 3307–3328 e19, https://doi.org/10.1016/j.cell.2022.07.016.
[28] N. Kundu, C.C. Domingues, J. Patel, M. Aljishi, N. Ahmadi, M. Fakhri, et al., Sucralose promotes accumulation of reactive oxygen species (ROS) and adipogenesis in mesenchymal stromal cells, Stem Cell Res. Ther. 11 (1) (2020) 250, https://doi.org/10.1186/s13287-020-01753-0.
[29] A.C. Sylvetsky, S. Sen, P. Merkel, F. Dore, D.B. Stern, C.J. Henry, et al., Consumption of diet soda sweetened with sucralose and acesulfame-potassium alters inflammatory transcriptome pathways in females with overweight and obesity, Mol. Nutr. Food Res. 64 (11) (2020) e1901166, https://doi.org/10.1002/mnfr.201901166.
[30] X. Bian, L. Chi, B. Gao, P. Tu, H. Ru, K. Lu, Gut microbiome response to sucralose and its potential role in inducing liver inflammation in mice, Front. Physiol. 8 (2017) 487, https://doi.org/10.3389/fphys.2017.00487.
[31] A. Pradhan, Obesity, metabolic syndrome, and type 2 diabetes: inflammatory basis of glucose metabolic disorders, Nutr. Rev. 65 (12 Pt 2) (2007) S152–156.
[32] A.M. Minihane, S. Vinoy, W.R. Russell, A. Baka, H.M. Roche, K.M. Tuohy, et al., Low-grade inflammation, diet composition and health: current research evidence and its translation, Br. J. Nutr. 114 (7) (2015) 999–1012, https://doi.org/10.1017/S0007114515002093.
[33] F.R. Greten, S.I. Grivennikov, Inflammation and cancer: triggers, mechanisms, and consequences, Immunity 51 (1) (2019) 27–41, https://doi.org/10.1016/j.immuni.2019.06.025.
[34] S Sen, et al., Sucralose Promotes Increased Fat Accumulation in Cultured Human Adipose Tissue Derived Mesenchymal Stem Cells, in: Energy Balance: Control of Adiposity and Feeding, 2015. SAT-556-SAT-.
[35] A. Shil, O. Olusanya, Z. Ghufoor, B. Forson, J. Marks, H. Chichger, Artificial sweeteners disrupt tight junctions and barrier function in the intestinal epithelium through activation of the sweet taste receptor, T1R3, Nutrients 12 (6) (2020) 1862, https://doi.org/10.3390/nu12061862.
[36] A.R. Basson, A. Rodriguez-Palacios, F. Cominelli, Artificial sweeteners: history and new concepts on inflammation, Front. Nutr. 8 (2021) 746247, https://doi.org/10.3389/fnut.2021.746247.
[37] V. Harrington, L. Lau, A. Crits-Christoph, J. Suez, Interactions of non-nutritive artificial sweeteners with the microbiome in metabolic syndrome, Immunometabolism 4 (2) (2022) e220012, https://doi.org/10.20900/immunometab20220012.
[38] M. Sanchez-Tapia, A.W. Miller, O. Granados-Portillo, A.R. Tovar, N. Torres, The development of metabolic endotoxemia is dependent on the type of sweetener and the presence of saturated fat in the diet, Gut Microbes 12 (1) (2020) 1801301, https://doi.org/10.1080/19490976.2020.1801301.
[39] A.V. Patel, E.J. Jacobs, D.M. Dudas, P.J. Briggs, C.J. Lichtman, E.B. Bain, et al., The American Cancer Society's Cancer Prevention Study 3 (CPS-3): Recruitment, study design, and baseline characteristics, Cancer 123 (11) (2017) 2014–2024, https://doi.org/10.1002/cncr.30561.
[40] Y. Wang, R.A. Hodge, V.L. Stevens, T.J. Hartman, M.L. McCullough, Identification and reproducibility of plasma metabolomic biomarkers of habitual food intake in a US diet validation study, Metabolites 10 (10) (2020) 382, https://doi.org/10.3390/metabo10100382.
[41] Nutrition Coordinating Center UoM, NDSR Software. [Internet] [cited April 24th, 2025]. Available from: 2019 https://www.ncc.umn.edu/products/.
[42] Y. Wang, R.A. Hodge, V.L. Stevens, T.J. Hartman, M.L. McCullough, Identification and reproducibility of urinary metabolomic biomarkers of habitual food intake in a cross-sectional analysis of the cancer prevention study-3 diet assessment sub-study, Metabolites 11 (4) (2021) 248, https://doi.org/10.3390/metabo11040248.
[43] J.M. Hodge, R. Shah, M.L. McCullough, S.M. Gapstur, A.V. Patel, Validation of self-reported height and weight in a large, nationwide cohort of U.S. adults, PLoS One 15 (4) (2020) e0231229, https://doi.org/10.1371/journal.pone.0231229.
[44] A.D. Nichol, C. Salame, K.I. Rother, M.Y. Pepino, Effects of sucralose ingestion versus sucralose taste on metabolic responses to an oral glucose tolerance test in participants with normal weight and obesity: a randomized crossover trial, Nutrients 12 (1) (2019) 29, https://doi.org/10.3390/nu12010029.
[45] A.G. Yunker, J.M. Alves, S. Luo, B. Angelo, A. DeFendis, T.A. Pickering, et al., Obesity and sex-related associations with differential effects of sucralose vs sucrose on appetite and reward processing: a randomized crossover trial, JAMA Netw. Open 4 (9) (2021) e2126313, https://doi.org/10.1001/jamanetworkopen.2021.26313.
[46] X. Bian, P. Tu, L. Chi, B. Gao, H. Ru, K. Lu, Saccharin induced liver inflammation in mice by altering the gut microbiota and its metabolic functions, Food Chem. Toxicol. 107 (Pt B) (2017) 530–539, https://doi.org/10.1016/j.fct.2017.04.045.
[47] A.C. Sylvetsky, R.A. Clement, N. Stearrett, N.T. Issa, F.J. Dore, R. Mazumder, et al., Consumption of sucralose- and acesulfame-potassium-containing diet soda alters the relative abundance of microbial taxa at the species level: findings of two pilot studies, Appl. Physiol. Nutr. Metab. 49 (1) (2024) 125–134, https://doi.org/10.1139/apnm-2022-0471.
[48] Y. Hanawa, M. Higashiyama, C. Kurihara, R. Tanemoto, S. Ito, A. Mizoguchi, et al., Acesulfame potassium induces dysbiosis and intestinal injury with enhanced lymphocyte migration to intestinal mucosa, J. Gastroenterol. Hepatol. 36 (11) (2021) 3140–3148, https://doi.org/10.1111/jgh.15654.
[49] X. Li, Y. Liu, Y. Wang, X. Li, X. Liu, M. Guo, et al., Sucralose promotes colitis-associated colorectal cancer risk in a murine model along with changes in microbiota, Front. Oncol. 10 (2020) 710, https://doi.org/10.3389/fonc.2020.00710.
[50] L.A. Mendez-Garcia, N. Bueno-Hernandez, M.A. Cid-Soto, K.L. De Leon, V.M. Mendoza-Martinez, A.J. Espinosa-Flores, et al., Ten-week sucralose consumption induces gut dysbiosis and altered glucose and insulin levels in healthy young adults, Microorganisms 10 (2) (2022) 434, https://doi.org/10.3390/microorganisms10020434.
[51] A.C. Sylvetsky, R.J. Brown, J.E. Blau, M. Walter, K.I. Rother, Hormonal responses to non-nutritive sweeteners in water and diet soda, Nutr. Metab. (Lond). 13 (2016) 71, https://doi.org/10.1186/s12986-016-0129-3.
[52] J.R. Dalenberg, B.P. Patel, R. Denis, M.G. Veldhuizen, Y. Nakamura, P.C. Vinke, et al., Short-term consumption of sucralose with, but not without, carbohydrate impairs neural and metabolic sensitivity to sugar in humans, Cell Metab. 31 (3) (2020) 493–502 e7, https://doi.org/10.1016/j.cmet.2020.01.014.
[53] A. Rodriguez-Palacios, A. Harding, P. Menghini, C. Himmelman, M. Retuerto, K.P. Nickerson, et al., The artificial sweetener Splenda promotes gut proteobacteria, dysbiosis, and myeloperoxidase reactivity, Crohn's disease-like ileitis, Inflamm, Bowel Dis 24 (5) (2018) 1005–1020, https://doi.org/10.1093/ibd/izy060.
[54] V.E. Hedrick, C. Nieto, M.F. Grilo, A.C. Sylvetsky, Non-sugar sweeteners: helpful or harmful? The challenge of developing intake recommendations with the available research, BMJ 383 (2023) e075293, https://doi.org/10.1136/bmj-2023-075293.